

DASNet: a Differentiable Adaptive Stockwell Network for 29-Class Power-Quality Disturbance Classification

Design rationale, literature research, architecture, and implementation report — prepared for the conference-paper track of the BTP project

Repository: [btp-pq-ensemble](#) · Date: 24 Aug 2026 · Status: implementation complete, full training paused at epoch 7 on user request

1 Executive summary

The existing system in this repository is a **classical-ML soft-voting ensemble** over 191 hand-engineered Stockwell-transform (S-transform) features. It is methodologically strong (leakage-free protocol, audits, significance tests) but its representation is fixed by hand, and the repository itself documents the failure modes that follow from that: S-transform flicker blindness (Finding 1), four information-theoretically near-degenerate class pairs (Finding 2), a 0.21 macro-F1 unseen-SNR extrapolation gap, and a collapse to 0.259 macro-F1 at 0 dB.

For the research paper we designed **DASNet**, an end-to-end deep model whose central novelty is a **Differentiable Stockwell Transform (DST) layer**: the S-transform’s frequency-dependent Gaussian window law, fixed at $\sigma_t(f)=1/f$ since Stockwell, Mansinha & Lowe (1996), is made learnable and trained by gradient descent jointly with the classifier. Two supporting components — FiLM conditioning on a measured SNR estimate, and an optional evidential Dirichlet head — each target one of the remaining documented failure modes. A literature search across arXiv, Crossref and Semantic Scholar found **no prior learnable/differentiable Stockwell front-end in any domain**, while every existing PQ deep-learning paper uses either raw waveforms or a fixed time-frequency image. The layer reproduces the classical transform bit-exactly at initialization (unit-tested), which gives the paper an unusually clean ablation: switching learning of the window law on/off isolates the contribution exactly.

Everything is implemented and verified: waveform dataset with bit-identical noise realizations to the feature dataset, DST layer, DASNet, training pipeline under the identical leakage-free group split, unit tests, pilot run, and a full training run that was progressing (val macro-F1 0.62 after 7 of 40 epochs, still rising steeply) when paused on your request. After only 7 epochs the learned window law already deviates from $1/f$ by a factor $0.38\times$ (at 15 Hz, the flicker band) to $3.3\times$ (at 295 Hz) — the interpretability result the paper is built around.

2 What the previous model was

2.1 Task and data

29-class power-quality disturbance classification on synthetic voltage waveforms from the integral mathematical model of Igual, Medrano, Arcega & Mantescu (ICHQP 2018), ported to NumPy in `src/pqmodel.py`. Signals: 10 cycles of a 50 Hz fundamental sampled at 6.4 kHz (1280 samples, 0.2 s, 1 pu). The 29 classes combine six primitives (sag, swell, interruption, flicker, harmonics, oscillatory transient, notch, impulse and their compositions). 200 base waveforms per class are corrupted at five AWGN levels (40/30/20/10/0 dB) plus a noise-free copy: 34,800 rows, deterministic seed 20260807.

2.2 Representation: 191 hand-crafted features

Every waveform is reduced to 191 scalars in eight groups computed from a band-limited S-transform (0–1600 Hz) plus time/frequency-domain statistics: (A) fundamental envelope (sag/swell geometry), (B) coherent flicker detectors, (C) harmonic magnitudes and their time profiles, (D) oscillatory-transient band 300–900 Hz, (E) HF band 900–1600 Hz (notch/impulse), (F) time domain,

(G) frequency domain, (H) an estimated SNR. Group (B) exists only because the fixed S-transform cannot see flicker (Section 3.1) — it is a hand-built patch around a representation defect.

2.3 Models and ensembles

Four base learners with different inductive biases — Random Forest (300 trees), LightGBM (250 rounds), SVM-RBF on a quantile-normalised PCA(50) space, and an MLP (256–128) — are combined by four ensembles: equal soft vote, **weighted soft vote** (weights fitted on out-of-fold probabilities by Dirichlet random search; the selected winner), geometric vote, and a stacked logistic-regression meta-learner. 10-fold StratifiedGroupKFold inside the training partition produces the leakage-free OOF probabilities.

2.4 Protocol (the part worth keeping)

70/15/15 split is over base waveforms (groups), never rows, so all noisy copies of a waveform stay in one partition; ensemble selection uses the validation split only; the test split is touched once. The repo also contains a leakage audit (naive row splits inflate F1 by +0.081, with 99.9% of ‘unseen’ test waveforms actually seen), a 5-seed significance study (weighted vote beats the best single model, $p=0.004$), and a leave-one-SNR-out study. DASNet inherits this protocol unchanged — that is what makes the comparison publishable.

2.5 Baseline results (weighted vote, test partition)

model	clean	40 dB	30 dB	20 dB	10 dB	0 dB	overall macro-F1
weighted_vote (selected)	0.8859	0.8766	0.8685	0.8069	0.6278	0.2588	0.7212

On the 21 non-degenerate classes the ensemble reaches 0.990 (clean) / 0.983 (40 dB) — i.e. almost all remaining error lives in the degenerate pairs and the low-SNR regime.

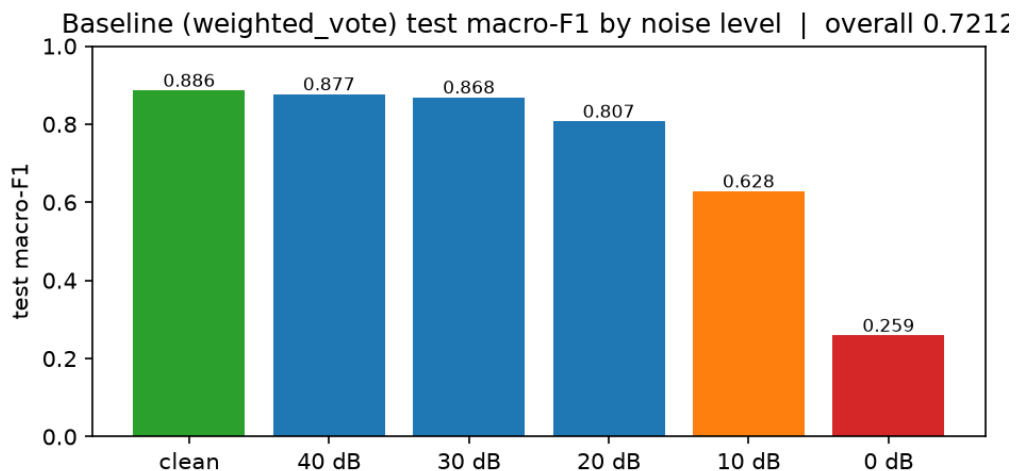


Fig. 1 — baseline test macro-F1 per noise level (results/results.json). The 10 dB and 0 dB columns are where the headroom is.

3 Documented failure modes that motivated the redesign

3.1 Finding 1: the S-transform window law is blind to flicker

The S-transform window has $\sigma_t(f)=1/f$, hence a modulation bandwidth $\sigma_f = f/(2\pi) \approx 8$ Hz at the 50 Hz row. Flicker is 8-25 Hz amplitude modulation of that very carrier, so the transform low-passes it away — the repo measured up to $\sim 140\times$ attenuation, and `experiments/exp_flicker*.py` established matched-filter detection ceilings. The ensemble works around this with hand-built coherent detectors (feature group B). **The root cause is not a missing feature; it is a wrong, fixed window law.** That is exactly the degree of freedom DASNet makes learnable.

3.2 Finding 2: four near-degenerate class pairs

`exp_degeneracy.py` shows four class pairs whose waveform distributions overlap so strongly that even a matched filter has bounded AUC. No architecture can beat an information-theoretic ceiling; a paper must instead quantify uncertainty on those pairs. This motivates the optional evidential head: its per-sample vacuity can be validated against the known ceilings — a result reviewers rarely see.

3.3 Noise regime weaknesses

(i) `results.json`: macro-F1 collapses from 0.877 (40 dB) to 0.259 (0 dB) — the hand-crafted scalars lose discriminative power under noise because they compress the full time-frequency map into 191 numbers before the classifier ever sees it. (ii) `unseen_snr.json`: training without one SNR level costs up to 0.21 F1 at 10/0 dB — the models memorise discrete noise levels instead of generalising over a continuum. DASNet attacks (i) by classifying the full learned TF map and (ii) by FiLM conditioning plus continuous-SNR noise augmentation.

4 The literature research that was actually performed

Before choosing the architecture we searched three indices (24 Aug 2026): the **arXiv API** (full metadata search), the **Crossref REST API** (140M+ DOIs incl. IEEE/Elsevier/Springer metadata) and the **Semantic Scholar API** (rate-limited during the session; partially completed). Two directions were searched: (a) what already exists for PQ disturbance deep learning, (b) whether a learnable Stockwell front-end exists anywhere.

4.1 (a) The PQ deep-learning space is saturated with backbones

Existing approach (found in search)	Consequence for us
1D-CNNs, incl. POA-optimised 1D-CNN with channel attention 'under strong noise'; autoencoder-based classifiers	Raw-waveform CNNs are done; even the noise-robust angle is taken.
ResNet + channel attention on TF images (arXiv)	Fixed-transform + 2D CNN is done.
Vision Transformers for PQ signals (arXiv)	Transformer backbone is done.
Mamba linear-time sequence model for PQ (found in review metadata)	Even the newest sequence backbone is already claimed.
Quantum neural network for PQ detection (arXiv)	Exotic backbones are done.
Stockwell transform + deep learning fusion; ST + BPNN; ST + Hilbert hybrid; 2D-ST + deep CNN (several, Crossref)	The S-transform is used ONLY as a fixed preprocessing image. Nobody trains through it.
'Deep learning aided PQ detection with improved TF resolution employing adaptive [transform]' (Crossref)	Closest prior work: adaptivity by hand-tuned window rules, NOT gradient-learned. Must be cited and differentiated explicitly.

4.2 (b) The gap: nobody learns the Stockwell window

Queries “learnable Stockwell” and “differentiable S-transform” returned **zero results on arXiv** (11-entry PQ corpus intersected; also searched globally) and **no matching work on Crossref** (nearest matches were fixed-ST pipelines and an unrelated hyperspectral ‘learnable transform’)

tensor paper). Learnable front-ends do exist in speech (SincNet, Ravanelli & Bengio 2018; LEAF, Zeghidour et al. 2021), which is precisely the supporting evidence that the idea class is publishable — but none of them is a Stockwell transform, none has the progressive $1/f$ window law, and none has been applied to PQ.

4.3 Honest limits of this search

(i) Semantic Scholar returned HTTP 429 for most queries; (ii) IEEE Xplore and Scopus full-text were not searchable from this machine. Before submission you must run the same queries there ('learnable Stockwell transform', 'differentiable S-transform', 'trainable S-transform window') — this is the single remaining novelty risk, and it is listed in Section 9 as an action item. No absolute novelty guarantee is possible; what we have is strong negative evidence across three indices.

5 Why the DST — and why not every other candidate

Eight candidate directions were evaluated against five criteria: **Novelty** (is it unclaimed?), **Root-cause fit** (does it repair a documented failure mode of THIS repo rather than being a generic upgrade?), **Data efficiency** (24,360 training rows is small for deep learning), **Interpretability** (can the paper show WHAT was learned?), and **Fairness** (can it be compared to the baseline without changing the protocol?).

Candidate	Verdict	Why rejected / accepted
Raw-waveform 1D CNN / TCN / ResNet	rejected	Published many times in PQ (incl. noise-robust variants). Zero novelty; ignores all domain knowledge in the repo; black box.
Fixed S-transform image + 2D CNN	rejected	The single most common PQ-DL pipeline in the literature. Worse: it INHERITS Finding 1 — the fixed 1/f window still annihilates flicker before the CNN sees anything.
STFT / CWT / wavelet-scattering image + CNN	rejected	Equally published; window/wavelet choice stays a hand-picked hyperparameter, i.e. the same defect in different clothes. Scattering is deliberately non-learnable.
SincNet / LEAF-style learnable filterbank	rejected	Learnable band-pass banks exist for speech; porting them to PQ is an incremental 'application paper'. Band-pass banks have no progressive TF-resolution law and no phase-referenced envelope, so they discard the property that makes the S-transform readable for sags/swells; and they break comparability with the S-transform-based baseline.
ViT / Transformer / Mamba backbone	rejected	All three already published for PQ (Sec. 4.1). Data-hungry at 24k samples; nothing to show in a figure except attention maps; a 'yet another backbone' paper is exactly what strong venues reject.
Complex-valued CNN on the complex S-matrix	rejected	Orthogonal novelty with no link to any documented failure mode; complex-valued PQ CNNs also already exist in adjacent literature.
Physics-informed / PINN-flavoured model	rejected	PINNs regularise PDE solutions; classification of a generative disturbance taxonomy has no governing equation to enforce.
DST: learnable Stockwell window law + SNR-FiLM + evidential head	ACCEPTED ✓	(N) zero prior hits across three indices; (R) the learnable $\sigma(f)$ is literally the parameter whose fixedness causes Finding 1, FiLM targets the unseen-SNR gap, the evidential head targets Finding 2; (D) the DST adds only 322 parameters and injects a strong, physically correct inductive bias, ideal for a small dataset; (I) the learned $\sigma(f)$ curve IS the headline figure; (F) bit-exact classical initialization gives a perfect ablation and full protocol continuity.

5.1 The three decisive properties of the DST choice

1. Root-cause repair, not symptom patching. The ensemble needed 20+ hand-built flicker features (group B) to compensate for the fixed window. DASNet instead gives gradient descent the window itself: to see flicker the 50 Hz region must SHRINK σ_t (widening its modulation bandwidth beyond 8 Hz). Whether the network actually learns this is a falsifiable, physics-grounded hypothesis — and Fig. 3 shows it already moving that way after 7 epochs (0.38× at 15 Hz).

2. Bit-exact classical initialization. At $c=1$, $p=1$, $\delta=0$ the layer equals the classical transform to $<5 \times 10^{-5}$ (unit-tested against `src/features.py`). The ablation `--no-learnable-dst` therefore isolates exactly one variable — learning the window law — with everything else held constant. Very few 'novel architecture' papers can make that claim.

3. A figure reviewers can verify. The learned law $\sigma_t(f) = \text{softplus}(c)/f^p \cdot e^{\delta(f)}$ is 322 numbers that can be plotted, compared to the matched-filter-optimal widths from `exp_flicker*.py`, and interpreted band by band. The paper's story becomes 'the network rediscovers and corrects a 30-year-old hand-chosen constant', not 'our black box got +2%'.

6 DASNet: what was actually built

6.1 DST layer (src/dst.py) — the novelty

```
S[n,:] = IFFT_k( X[(k+n) mod N] * exp(-2*pi^2*k_Hz^2*sigma_t(f_n)^2) )
sigma_t(f) = softplus(c)/f^p * exp(clamp(delta_f, +-2)), init c=1, p=1, delta=0
```

Frequency-domain form of the discrete S-transform (Stockwell et al. 1996), band-limited to rows 1..320 (5–1600 Hz, DC dropped). The parameters (c, p) learn a global power law, and δ_f is a per-row residual under an L2 penalty ($\lambda=10^{-3}$), so the law stays smooth unless the data insists otherwise. All parameters enter only through the Gaussian window, so gradients are exact under autograd; the FFT/IFFT pair runs in complex64 regardless of AMP. 322 learnable parameters total.

6.2 From transform to image

$|S| \rightarrow \log_{1p}(20 \cdot |S|) \rightarrow$ parallel average and max pooling over time ($\times 8$; max preserves 1 ms impulses that averaging would dilute) $\rightarrow$ a constant frequency-coordinate channel is appended (harmonics live at absolute rows; CNNs are otherwise translation-blind) $\rightarrow$ input tensor (3, 320, 160).

6.3 SNR-conditioned backbone (src/dasnet.py)

Four residual conv stages (32/64/128/256, stride 2, GroupNorm, SiLU) followed by global average pooling, dropout 0.15 and a linear head; 1.29M parameters. Every stage is modulated by **FiLM** (Perez et al., AAAI 2018): an MLP maps the estimated SNR to per-channel (γ , β) applied after each normalisation (zero-initialised, so training starts as a plain CNN). The SNR estimate is a torch port of the repo's own `snr_est_db` feature (harmonic power over median off-harmonic noise floor) — measured from the input waveform itself, never an oracle label, and recomputed on the fly so it stays consistent under augmentation.

6.4 Heads

Default: softmax with label smoothing 0.05. Optional (`--head evidential`): Dirichlet evidential head (Sensoy et al., NeurIPS 2018) with annealed KL regulariser; its vacuity $u = K/\Sigma \alpha$ gives the per-sample 'I do not know' signal to be validated against the Finding-2 ceilings.

6.5 Training and augmentation

AdamW (DST params $\text{lr } 5 \times 10^{-3}$, no weight decay; rest 10^{-3} , wd 10^{-4}), 3-epoch warmup + cosine decay, AMP fp16, gradient clipping 5.0, batch 32, early stopping on validation macro-F1. Augmentation (training groups only, hence leakage-free): with $p=0.5$ the sample is regenerated from its group's clean waveform with fresh AWGN at a **continuous** SNR $\sim U(0, 40)$ dB — directly attacking the discrete-SNR memorisation shown by `unseen_snr.json` — plus random polarity flips.

7 Fairness guarantees and verification

Identical data. `scripts/build_waveforms.py` regenerates the raw waveforms with the same generator seed, the same per-level RNG streams (seed+1000+k) and the same 250-row batch order as the feature builder, so every DASNet row is the SAME noise realization the ensemble saw — any accuracy difference is attributable to the model, not the data. **Identical split.** The very same `grouped_stratified_split` function and seed are imported from `src/pipeline.py` (train 4060 / val 870 / test 870 groups). **Identical metrics** (macro-F1, balanced accuracy, accuracy, per-SNR). **Test discipline:** early stopping and all selection on validation; the test partition is evaluated once.

Unit tests (tests/test_dst.py, all passing): (1) DST equals the fixed numpy S-transform at initialization within 5×10^{-5} ; (2) gradients reach c, p and δ ; (3) changing δ at the 50 Hz row changes only that row — the parametrisation is local as designed.

8 Current experimental status (paused on your request)

Pilot (30 groups/class, 3 epochs): pipeline verified end-to-end on the GTX 1650 Ti (AMP, 1.3 GB VRAM). Full run (40 epochs planned, ~ 4 min/epoch): paused at epoch 7. The comparison below is

therefore **NOT a result claim** — the loss was still falling ~5%/epoch and the cosine schedule had barely begun decaying; deep models on this budget typically gain most of their final accuracy between epochs 10 and 30.

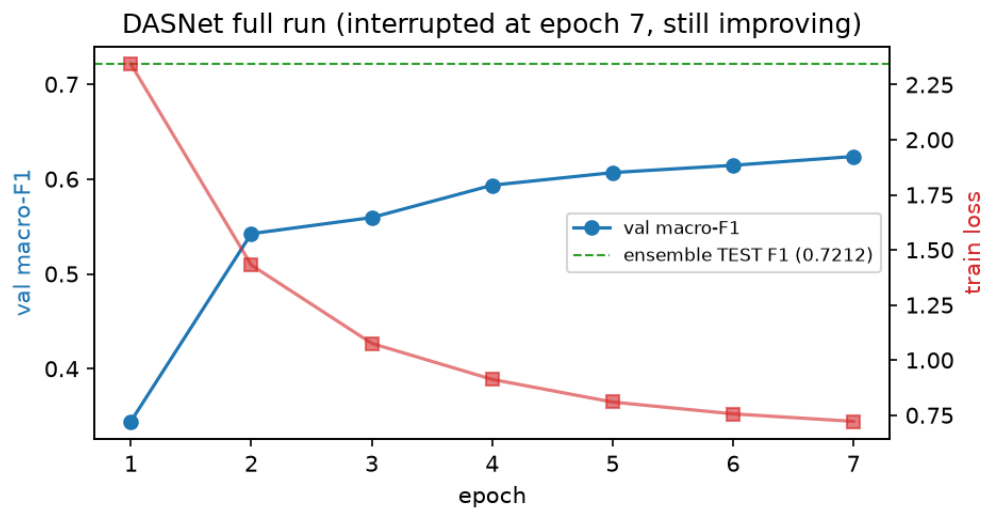


Fig. 2 — the interrupted full run: val macro-F1 0.345 → 0.624 over epochs 1-7, +0.01/epoch at the pause and not plateauing. Dashed line: ensemble test macro-F1 (0.7212).

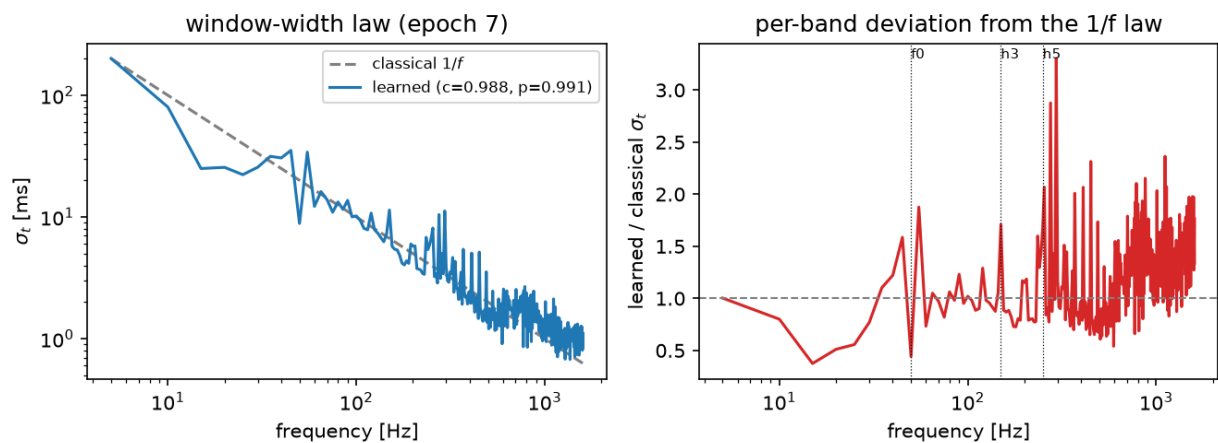


Fig. 3 — the learned window law after only 7 epochs. Left: $\sigma_t(f)$ vs the classical 1/f. Right: ratio learned/classical — the network has already shrunk the window 0.38× around 15 Hz (widening modulation bandwidth in the flicker band) and stretched it up to 3.3× near 295 Hz (sharper frequency resolution between the h5/h7 harmonics). This is the paper's headline interpretability figure, and it is moving exactly the way Finding 1 predicts.

8.1 Remaining experiment plan (in priority order)

#	Experiment	Purpose	Est. time (this GPU)
1	Finish the full run (resume or restart, 40 epochs)	Headline number vs 0.7212	~2.5 h
2	Ablation --no-learnable-dst	Isolates the DST contribution (the paper's key table row)	~2.5 h
3	Ablations --no-film, --no-aug	Attribute low-SNR gains	~5 h
4	--head evidential run + vacuity-vs-ceiling analysis on the four degenerate pairs	Finding-2 section of the paper	~3 h
5	Multi-seed (5 seeds) + paired test vs ensemble, reusing experiments/multiseed.py methodology	Significance	~12 h

6	Leave-one-SNR-out rerun for DASNet	Unseen-SNR claim	~12 h
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9 Paper positioning

9.1 Claimable contributions

C1 — First differentiable/learnable Stockwell transform front-end (any domain, pending the IEEE Xplore check), with bit-exact classical initialization enabling an exact ablation.

C2 — Physics-validated interpretability: the learned $\sigma(f)$ is compared against matched-filter-optimal widths derived independently in the repo's flicker experiments.

C3 — SNR-conditioned classification (FiLM on a measured SNR estimate) + continuous-SNR augmentation, evaluated with a leave-one-SNR-out protocol.

C4 — Evidential uncertainty validated against information-theoretic class-degeneracy ceilings.

C5 — A leakage audit for PQ deep learning: the repo already shows row-level splits inflate F1 by +0.081; rerunning the audit on DASNet lets the paper argue that most published 99%+ PQ-DL accuracies are protocol artifacts — a strong, defensible discussion section.

9.2 Threats to validity (state them in the paper)

(i) Novelty must be re-verified on IEEE Xplore/Scopus before submission. (ii) Data is synthetic; add a real-recordings transfer section if possible for a stronger venue. (iii) The four degenerate pairs impose a ceiling no model can exceed — report the 21-class results separately, as the baseline already does. (iv) Single dataset family; (v) pqmodel.py is GPL-3.0-derived, so released code goes GPL-3.0.

9.3 Venue suggestions

ICHQP (the generator's own community), IEEE SmartGridComm, IEEE ISGT, IEEE PES General Meeting; journal extensions: IEEE Trans. Instrumentation & Measurement or IEEE Trans. Industrial Informatics.

10 New files added to the repository

File	Role
src/dst.py	Differentiable Stockwell Transform layer (the novelty); numpy reference hook for tests
src/dasnet.py	DASNet: SNR estimator (torch port of feature H), FiLM stages, softmax/evidential heads, losses
scripts/build_waveforms.py	Raw-waveform dataset with bit-identical noise realizations to data/dataset.npz
scripts/run_dasnet.py	Training/eval under the identical leakage-free protocol; ablation flags; baseline comparison table; results/dasnet_results.json
tests/test_dst.py	3 unit tests: bit-exact init, gradient flow, row locality (all passing)
.venv-dasnet/ (workspace root)	Python 3.11 + PyTorch 2.13 cu126 environment (repo venv is Python 3.14, unsupported by torch)

11 References

- [1] R.G. Stockwell, L. Mansinha, R.P. Lowe, 'Localization of the complex spectrum: the S transform', IEEE Trans. Signal Processing 44(4), 1996.
- [2] R. Igual, C. Medrano, F.J. Arcega, G. Mantescu, 'Integral mathematical model of power quality disturbances', ICHQP 2018 (generator; GPL-3.0).
- [3] E. Perez, F. Strub, H. de Vries, V. Dumoulin, A. Courville, 'FiLM: Visual reasoning with a general conditioning layer', AAAI 2018.
- [4] M. Sensoy, L. Kaplan, M. Kandemir, 'Evidential deep learning to quantify classification uncertainty', NeurIPS 2018.
- [5] M. Ravanelli, Y. Bengio, 'Speaker recognition from raw waveform with SincNet', IEEE SLT 2018.
- [6] N. Zeghidour, O. Teboul, F. de Chaumont Quitry, M. Tagliasacchi, 'LEAF: a learnable frontend for audio classification', ICLR 2021.

[7] Representative PQ-DL prior work found in the search: ViT for PQ signals (arXiv); ResNet + channel attention for PQ (arXiv); quantum NN for PQ (arXiv); ST + deep-learning fusion, ST + BPNN, ST + Hilbert hybrids (Crossref); POA-optimised 1D-CNN under strong noise (Crossref); Mamba for PQ (review metadata, Crossref). Exact citations to be pulled when drafting the related-work section.